

Elaine(Yilin) Cai

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EDUCATION

University of California, San Diego

B.S. in Computer Science | GPA: 3.5/4.0

San Diego, CA

Sep 2024 - June 2027 (expected)

Pasadena City College

Major in Computer Science | GPA: 3.8/4.0

Pasadena, CA

Aug 2022 - June 2024

EXPERIENCE

Software Engineering Intern – Nest Friends (Remote)

Nov 2025 – Jan 2026 (expected)

Nest Friends is an early-stage U.S. startup building a homeowner-focused dating app that recently launched on the App Store and is preparing for its version 2 release.

- Collaborated directly with founders and a small developer team to enhance the app's second release, focusing on UI/UX improvements, bug fixes, and feature expansion (face verification and safety modules).
- Contributed to frontend development using **React Native**, API integration with **MongoDB**, and deployment on **AWS / Google Cloud**.
- Helped stabilize and optimize the in-app user experience by improving data flow and component state management for profile and authentication features.
- Collaborated with the founders to test builds and document bugs through **Slack and JIRA**, ensuring effective communication across time zones with external teams.
- Gained first-hand experience in early-stage startup engineering, rapid iteration cycles, and cross-functional product decision-making.

PROJECTS

HashFlow LoadBalancer - Distributed Load Balancing System

October 2024

- Architected production-grade load balancer using consistent hashing algorithm achieving 10,000+ req/s throughput with <10ms latency and 99.9% uptime. Implemented RESTful API for dynamic server management, Docker containerization with health monitoring, and conducted performance testing with visualization dashboard (Python, Flask, Docker, pytest, Repository pattern).

Panda Foodies - Enterprise Full-Stack Web Application

September 2025

- Built full-stack food delivery platform with Spring Boot backend (15+ REST APIs, JWT/BCrypt security, PayPal integration, 75%+ test coverage) and React SPAs for customer/admin interfaces. Optimized MongoDB with compound indexes improving query performance 45%, deployed to AWS S3 with IAM, achieved OWASP compliance (Java, Spring Security, React, MongoDB, AWS, JUnit, Agile/Scrum).

JobSwipe - Full-Stack Job Discovery Platform with Automation

March 2025

- Developed job discovery platform with Selenium-powered automation reducing application time 90% (12 min to 45 sec) with 95% success rate. Built Node.js/Express backend with intelligent matching algorithm (85% user satisfaction), responsive swipe-based UI achieving Lighthouse scores >90, localStorage persistence supporting 200+ listings (Node.js, Selenium, React, HTML5/CSS3).

Multi-Agent AI Platform

June 2025

- Designed multi-agent conversational system with specialized agents for math solving, weather retrieval, and task management. Built modular Flask backend with LangGraph integration and React frontend, configured CORS and RESTful endpoints for multi-origin support (Flask, Python, LangGraph, LLM APIs, React).

TECHNICAL SKILLS

Languages & Tools:

Programming: Java, Python, JavaScript/Node.js, HTML5, CSS3, SQL, C++

Backend: Spring Boot, Express.js, Flask, RESTful APIs, Microservices

Frontend: React, JavaScript (ES6+), HTML5/CSS3, Bootstrap

Database: MongoDB, PostgreSQL, Database Optimization

DevOps: Docker, Docker Compose, AWS, Google Cloud, Microsoft Azure

Testing: JUnit, Jest, pytest, Selenium WebDriver, TDD

Tools: Git, Maven, npm, Postman, IntelliJ IDEA, VS Code

Practices: Agile/Scrum, Design Patterns (MVC, Repository, Factory), CI/CD

Core Competencies:

Distributed Systems | Full-Stack Software Development | Object-Oriented Programming | RESTful API Design | Software Architecture & Design Patterns | Load Balancing & Scalability | Automated Testing | Security Best Practices | Cloud Infrastructure | Problem-Solving & Debugging

CERTIFICATION

Microsoft Certified: Azure AI Fundamentals

November 2025

Credential ID: 6993FDEE1996238F